Riley Beenders

PRODUCTION SYSTEMS | COTS PLATFORM SUPPORT | INFRASTRUCTURE & INTEGRATION

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PROFESSIONAL SUMMARY

Engineer who keeps business-critical production systems and vendor-provided platforms running, from go-live readiness and stabilization to the root-cause fixes that stop a problem from coming back. Works across application, integration, network, and hardware layers, and explains each issue and its fix clearly to vendors, operators, and leadership.

CORE EXPERTISE

Production Support & Operations: Implementation and go-live; post-implementation stabilization; root cause analysis; troubleshooting across compute, networking, storage, and system integrations; technical escalation point; vendor coordination and resolution; operational documentation and user guides; risk analysis and mitigation; data migration and inventory reconciliation

Vendor & COTS Platforms: Microsoft Dynamics 365 Business Central (cloud-hosted on Azure); SimplyPrint; Ubiquiti UniFi; Siemens PLCs; Duet3D controllers; Mean Well power supplies; Allen Bradley PLCs

Infrastructure & Networking: Windows/Linux systems administration; Debian servers; Kubernetes; Docker and containerized applications; Proxmox virtualization; VLAN segmentation; DHCP and IP address management; DNS; reverse proxy and firewall routing; network monitoring; multi-site network management; Raspberry Pi device fleets

Software, Data & Delivery: Python (asyncio, Starlette); Modbus TCP; SQLite; Server-Sent Events; PowerShell and WinGet; TypeScript and Electron; Git and GitHub; LabVIEW; data logging and process analysis; technical writing; worldwide project collaboration

PROFESSIONAL EXPERIENCE

Proteor / Proteor Printing Solutions | Coplay, PA — January 2025 - Present

Lead, Research & Development Engineer

Proteor acquired Filament Innovations in January 2025, and the CTO role below continued as Lead R&D Engineer for the same product line, now Proteor Print.

- Took the Coplay location's Microsoft Dynamics 365 Business Central ERP, a cloud-hosted platform on Azure, from evaluation to live production in five weeks: two weeks evaluating the software, then three migrating more than 2,000 items and 150+ products into the live system, reconciling inventory counts on the floor, and syncing SKUs with Proteor's global locations.
- Led the 2026 inventory count across every item in Business Central, traced its discrepancies to production orders that were processed without verification and never took items out of inventory, and helped implement a more warehouse-focused workflow.
- Own ExtrusionStudio, the QC and production system live on all three filament extrusion lines since October 2026, which I took from a LabVIEW proof of concept to a Python service; architecture and results are under Selected Projects.
- Serve as the plant's technical subject-matter expert, from industrial electronics on the extrusion lines to low-voltage sensors, and the escalation point for customer issues that go beyond what support can resolve, across mechanical faults, electrical faults, and user error.
- Support the vendor platforms the plant and products run on, including the Siemens PLCs on the extrusion lines, Duet3D controllers, and Mean Well power supplies, working with vendors through issues that regularly run into nights and early mornings.
- Contributed the machine requirements and integration for ProteorPrintOS, built with an external software partner on SimplyPrint, giving users remote operation, live video, AI-assisted failure alerts, and material, waste, and outcome metrics.
- Managed and expanded the division's IT infrastructure after the acquisition, standardizing software and internal processes to align with Proteor's company-wide systems.
- Developing an offline version of the ICARUS-Lite and ICARUS-Pro control screen with over-the-air automated updates, so customers can run without a Wi-Fi or Ethernet connection.
- Present project status, constraints, and budgets at Proteor's monthly USA review with leadership from its global locations; estimate and track the budget of every engineering project, build its BOMs, and maintain each product's unit cost in Business Central.
- Performed risk analysis and mitigations to EU and U.S. standards across ProteorPrint hardware and cloud products.
- Owned the ICARUS-Lite, a compact industrial printer that automates prosthetic and orthotic production, through commercialization against unit goals of 60 in 2025 and 100 in 2026, leading its mechanical, electrical, and software design.

- Standardized POSEIDON and ICARUS on one linear-motor architecture with swappable mounts, reducing product-to-product complexity and simplifying service; validated 4-micron precision at 4 m/s with a 30 kg moving mass.
- Co-developed a custom hotend with an external manufacturing partner, improving reliability and reducing cost for the company's materials.
- Act as Hiring Manager for engineering roles, conducting technical interviews and owning final hiring decisions.

Filament Innovations | Whitehall, PA — May 2020 - January 2025

Across these roles, wrote customer proposals, scopes of work, and estimates, and began tracking project budgets.

Chief Technology Officer — January 2023 - January 2025

- Oversaw the product-development pipeline, internal IT infrastructure, and technology strategy while keeping hands-on responsibility for mechanical, electrical, firmware, and software systems, and gave the CEO and the company regular updates as designs and software iterated.
- Led the design and launch of Gen3 POSEIDON, the flagship's production redesign, combining a 1 m³ build volume, filament-and-pellet dual extrusion, and braked Z drives that prevent uncontrolled motion during power loss.
- Researched high-speed linear motors that moved a 30 kg payload at 3 m/s with 5-micron accuracy and conducted early 6-axis robotic non-planar printing research.

Lead Electro-Mechanical Engineer — June 2022 - January 2023

- Integrated a multi-board CAN-bus control architecture that simplified wiring and debugging on larger, modular machines.
- Improved thermal and acoustic performance through enclosure redesign and component placement, including custom tooling with local fabricators for large-radius aluminum-panel bends.
- Removed a manufacturing bottleneck with independently testable electronics enclosures that installed in minutes, coordinating custom harnesses, cut panels, and hardware improvements with vendors.

Industrial Engineer — April 2021 - June 2022

- Delivered THE KRAKEN, a 1 x 2 x 1.5 m hybrid pellet-and-filament gantry machine for a client producing the world's largest 3D-printed statue, carrying it from CAD, BOM, and sourcing through fabrication, assembly, and firmware tuning to customer support and operator training.
- Designed and released Kratos, a machine engineered to print high-strength above-knee and below-knee prosthetic sockets.
- Developed next-generation ICARUS and POSEIDON models with higher-performance components and expanded motion safety through NEMA 34 motors, planetary gearboxes, and power-failure brakes.

Engineering Technician — May 2020 - April 2021

- Diagnosed mechanical and electrical problems during assembly and operation, improving early product reliability.
- Tuned, calibrated, and optimized machine parameters while documenting design limitations and improvement opportunities.

ADDITIONAL RELEVANT EXPERIENCE

Northampton Area School District - Computer / Network Technician | June 2016 - August 2019 | Built network infrastructure and deployed more than 750 wireless access points across six school buildings; accelerated deployment of more than 2,500 Chromebooks with an Arduino-based tool that automated setup-data entry, and coordinated final device distribution with school administrators.

Blue Mountain Ski Resort - Ski Lift Operator (seasonal) | November 2019 - January 2021 | Operated lifts and assisted thousands of guests with ticketing, weather, and slope information; performed motor and control safety checks and supported the maintenance team during lift breakdowns.

RJB.Engineering - Owner & Engineering Consultant | 2015 - Present | Deliver project-based engineering support for automotive clients and operate a small online store for printer heat inserts and accessories.

Homeowners Association - Treasurer | April 2026 - Present | Manage the banking and monthly insurance payments for an association of about 80 units, and help establish a community-focused member board in place of a property management company, remaining Treasurer through a mid-year restructure.

EDUCATION & CERTIFICATIONS

Penn State University | B.S., Electro-Mechanical Engineering — December 2024

Machine Learning Onramp, MathWorks	2026
MATLAB Onramp, MathWorks	2026
Simulink Onramp, MathWorks	2026
Simscape Onramp, MathWorks	2026
Lean Six Sigma Yellow Belt, Educate 360	2025
Lean Six Sigma White Belt, Educate 360	2025
Certificate in Engineering Design and Tools, Penn State University	2023
Mechanical Engineering Certificate, Lafayette College	2019

SELECTED PROJECTS

ExtrusionStudio | Plant measurement and production system, Proteor — September 2025 - Present

Python, asyncio, Modbus TCP, Starlette, SQLite, Server-Sent Events, Raspberry Pi, Debian

Problem: Each of three extrusion lines had a tri-axial laser gauge, and the only record of a spool's diameter was whatever an operator wrote down. The LabVIEW proof of concept was one program compiled once per line with a hard-coded address and delay, so a line could not be added without a rebuild, a dead line held up the others, and no reading was tied to a spool.

- Runs one asyncio poller and Modbus TCP session per line, reading every line concurrently at 20 Hz so a powered-down line never stalls a healthy one; the poll rate is per-line configuration with a floor, because the right value depends on the PLC's response time.
- Added a calibration phase that times the filament from the laser to the spool, a delay of one to two minutes through the accumulator, so each reading and alarm lands on the spool it is actually wound on instead of firing at the laser.
- Logs every spool to its own CSV and a SQLite run table with a serial, QR label, and HTML certificate; the packing bench approves, inspects, or rejects each spool against the gauge's figures and prints spool, box, and master case labels on demand, removing up to about 30% of label waste per run.
- Runs a fleet of Raspberry Pi kiosks (projector, a touch screen per line, packing bench, statistics TV) that check in with the server and are updated, configured, and switched on and off with the work hours from one settings panel.
- Built for supportability: Server-Sent Events that reconnect by themselves, no internet dependency on the floor, guides served by the server itself, a hand-written Modbus simulator that shares no code with the client, and a server that updates itself daily from its checked-out git branch so a fix can be tried on the real system.

Unified UniFi Network | Network infrastructure, Proteor plus two residential sites — March 2024 - Present

Ubiquiti UniFi cloud gateways, VLANs, DHCP, DNS, firewall routing, network monitoring

Problem: As production grew, the plant's consumer mesh network assigned the same IP address to machines that came online together and never reliably detected or resolved the conflicts.

- Replaced it with a centrally managed UniFi network, using VLANs to separate production equipment, work devices, and IoT devices, and took central control of IP address assignment, DNS, VLANs, and port forwarding.
- Reduced IP conflicts and network-related downtime, improved security through segmentation, and gave room to add equipment; monitoring tools now surface connectivity problems so they are resolved faster, and all three sites are configured and monitored remotely from one interface.

Engineering Operations Platform | Internal ERP/CRM, Filament Innovations (as CTO)

Kubernetes, ERP/CRM development, BOM automation, infrastructure design

Problem: Orders, inventory, tasks, projects, customer interactions, engineering BOMs, and purchasing ran as separate manual workflows.

- Designed and hosted a redundant Kubernetes cluster running a self-developed ERP/CRM that centralized those records, generated BOMs dynamically, and linked design data directly to purchasing, reducing the separation between design decisions and purchasing execution.

BulkSSH | Windows desktop app, github.com/RileyBeenders/BulkSSH — October 2026 - Present

TypeScript, Electron, Node.js, SSH

Problem: Testing ExtrusionStudio's Pi kiosks meant repeating the same commands on every Pi, one SSH window at a time, after every change to the image.

- Opens many devices in a grid of live terminals, mirrors keystrokes to every ticked device, or broadcasts one command and collects each device's output and exit code, so an image change is tried on every Pi in one pass.
- Destructive commands such as reboot or rm -r need confirming before they reach several devices, host keys are never accepted silently, and dropped connections reconnect by themselves; released as an installer and a portable executable.

RB's Auto App Updater | Windows utility — September 2026

PowerShell, WinGet, Windows Task Scheduler, GitHub, SHA-256 verification

Problem: Windows applications update through different methods, so keeping dozens of them current meant visiting websites, downloading installers, and running each one by hand.

- Updates every WinGet-managed application with one double-click and no UAC prompt after a one-time setup, running the upgrades through an elevated scheduled task and upgrading each package on its own so a single failure never stops the rest.
- Translates WinGet exit codes into plain-language statuses such as security blocked, restart needed, and application in use, writes detailed logs that make remote troubleshooting possible, and updates itself only after the new script's SHA-256 hash matches its version manifest.